CREATION AND IMPLEMENTATION OF POLICY FOR INFORMATION TECHNOLOGY USE: ABOUNDING ETHICS AND VALUES ISSUES

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ABSTRACT

The University of Michigan recently instituted a university-wide policy to encourage ethical and responsible use of information resources. Ethics and values issues abounded making the process of developing the policy itself, lively and interesting. The paper presents an overview of issues, dilemmas and value conflicts faced during the development of the Michigan policy. Implementation however is the most essential measure of any policy's effectiveness. Therefore the paper also presents one model for policy implementation. The first year of implementation of this model has just been completed at Michigan with encouraging results. A summary of key steps for developing and implementing policy in this area are presented.

INTRODUCTION

Universities and colleges across the nation are incorporating information technology into the way in which they do business at a very rapid rate. Members of these communities are increasingly exploring and experimenting with the capabilities of the new technologies and of information resources in general, for their teaching, research and learning.

With the new technological capabilities to access, organize, alter, create, reproduce and store information, also comes the ability to complete acts before they are entirely thought through, before the implications of the transaction are examined for human effects, sometimes even before responsibility is considered. The speed of transactions and the new capabilities of the technology, in the hands of increasing numbers of users, has resulted in a number of behavioral incidents that cause institutional concern.

TECHNOLOGY USE AND INSTITUTIONAL DILEMNAS

On University and college campuses, the wide variety of behavioral incidents is also creating a set of ethical and values questions to be raised. These questions often center around the definition of responsible use of the technology, around responsibilities of institutions to individuals within their communities, and around responsibilities of individuals in communities to their institutions.

Dilemmas are created when two equally held values or equally balanced alternatives come into conflict. For example, how should institutions protect information from tampering and secure institutional records, and yet encourage open access to the institutions information resources? How should institutions protect the community from electronic harassment and yet ensure freedom

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of speech? Institutions have been thrust into this set of dilemmas without much preplanning. Naisbitt and Aberdene, in "Megatrends 2000" write:

"The responsibility for what is happening has already been thrust upon us. Technology is not inherently evil. It is neutral. How we use it is not neutral."

Conflict first arises when administrators receive a complaint and need to respond to a behavior but are unclear as to what is involved. Lack of clarity may come because the individual lacks familiarity with the technology (not understanding its capabilities) or because of the uniqueness of the particular incident, (a new application of the technology). Under these circumstances the individual cannot readily identify the ethics and value issues that are embedded within the incident. e.g. Is this stealing or is it not? Is this falsification not? As an example, the ease with which individuals are able to copy software has caused many people to overlook the essence of the process--the act of taking something for which you have no ownership. This happens because people are sometimes unfamiliar with the fact that such software is copyrighted, and at other times they are swayed by the ease and the cost savings and choose to ignore the ownership issues.

Conflict next arises creating an institutional dilemma, when two parts of the institution value opposite poles on a values continuum and the behavioral incident falls somewhere between the poles. For instance, the technology allows alteration of visual images without detection. Some members of a university community may value the preservation and validity of representations of reality for archival and historical purposes. They may view the alteration of a visual image as a serious act requiring punishment. At the same time others may value enhancements of photographic work or other visual images as an artistic improvement to the composition or clarity of the piece of work. Instead of punishment, praise would be a potential from this portion of the community.

When such incidents come to the attention of administrators within the institution, sooner or later a policy will be requested. Values are threatened. Guidelines are needed to assist the community in its decision-making. At the present time, many colleges and universities are developing policies to define "appropriate or responsible use of information technology", standards on which to judge individual behavior.

POLICY DEVELOPMENT MODEL--DEFINING PROPER USE

The perceived purpose of institutional policy is interesting in and of itself. some people, policy is meant as a control tool for unacceptable behavior--the policy statement identifies the unacceptable behaviors, defines the range of jurisdiction over which the policy has influence, and defines the steps that wil be taken in response to violators. Stating the "thou shalt nots" is an end in itself.

For others however, policy is a standard setting process; it is the formal commitment of the institution to a standard of behavior to which the community commits itself and toward which end the disciplinary actions are directed for those whose behaviors fall short. So policy may be a statement of "thou shalt nots" and the "if you do.....these will happens", or policy may be statement of the "expectations" and the "community behavioral agreements".

Whatever the purpose institutions embrace for creating policy, Universities are increasingly taking a stand in regards to information technology use and responsibility within educational communities. In 1978, Sissela Bok appealed to public and private institutions to take a stand, writing:

*"The social incentives to deceit are at present very powerful; the controls, often weak. Many individuals feel caught up in practices they cannot change. It would be wishful thinking therefore to expect individuals to bring about major changes in the collective practices of deceit by themselves. Public and private institutions, with their enormous power to affect personal choice, must help alter the existing pressures and incentives."*2

COMMUNITY INVOLVEMENT IN POLICY MAKING

Policy can go further than serving simply as a tool for controlling behaviors or setting standards. It can also be the tool through which other societal, developmental and institutional mission related goals are reached. It can be the vehicle that stimulates discussions, that motivates the community to become involved in open discussions, to determine where it stands on particular ethical issues, to set community standards and to define how it wants to behave as

a community. In institutions of higher education, a policy can be a dynamic document that not only provides the standards and the sanctions, but focuses on process, moving into learning and teaching and on to action and practice. Again Sissela Bok is helpful in our thinking. She writes:

> *"There is strong insistence now that business ought to have a code of ethics. The codes must be but the starting point for a broad inquiry in the ethical quandaries encountered at work."*3

One may ask, why would such broad inquiry, discussions and community involvement be important? At the heart of my answer is the belief that people are more likely to abide by the policy if they have an investment in it, if they believe in it and if they think it is important to the quality of their own life institution. Beyond that however, I believe that information technology, because of its newness, its capabilities, its unique forms, provides us with an opportunity to look at and to better understand a set of value issues that exist between us as members of a community. Clarification and understanding of these value issues can help a community to become closer-more coherent and congealed. It can advance the community member's appreciation of differences and its valuing of the diversity that exists within an institution.

While I do not see these ethical dilemmas arising from the use of information technology, as new, I do believed that they are difficult to see because they concealed by our lack of understanding or awe of the technology. This requires that we establish new clarity on the values we hold, especially those that may b compromised by the capabilities of technology and its applications. Information technology and the proliferation of its use in society may provide us with an opportunity to reach a new level of maturity relative to social responsibility o may be the ground on which we show our moral regression.

Michigan's Model for Policy Making and Community Involvement The University of Michigan experience is briefly addressed in the following material. Michigan administrators needed to face the dilemmas that were arising from information technology use and conflicting institutional values. They realized and valued the diversity within the institution (an institutional community of 55,000 faculty, staff and students). And they sought to open discussions through community involvement and debate.

Key steps in policy development and implementation to date have been: 1) establishing need, 2) increasing awareness among decision makers of current behavioral incidents, 3) defining the role and expectations of policy, 4) clarify and debating values and community standards, 5) gaining high level administrative endorsement and promotion, 6) expanding policy draft to include information resources as well as information technology resources and networks, 7) executive officer approval, 8) information dissemination regarding the policy, 9) "Think About It" Campaign to stimulate awareness, debate, discussion, and ownership of ethics and values standards.

1. Establishing Need

In 1987, at the request of the Vice Provost for Information Technology and in response to an incident of harassment on one of the student electronic conferences, a process was undertaken to establish a policy to define proper use the information technology resources at Michigan. A subcommittee of the University Policy committee was formed. After months of discussion, they concluded that the University:

- is committed to the existence of diversity in its scholarly community,
- wishes to promote tolerance and respect for that diversity,
- wishes to promote and encourage open exchange of opinion and discussion of different points-of-view as an integral part of promoting maturation and learning within its community,
- wishes to promote the use of information technology and its potential for increasing the speed and breadth of communication and information exchange within its community,
- should ensure that its community is systematically and routinely made are of the ethical issues involving information technology usage through statements that guide the conditions of such use, and,
- should put in place guidelines to ensure that organizers of groups of users continue to focus attention on the values of the university and their implications for the use of

information technology. No policy was written. However some critical premises were established for both policy and process.

In 1988, the University's Information Technology Policy Committee requested that a policy be written to define proper use of information technology at the University of Michigan and to describe for the community the kind of open environment of information sharing that was being built, highlighting the user responsibilities that accompany such an information environment.

2. *Increasing Awareness of Incidents*

A collection of vignettes regarding incidents that were occurring on the campus was begun. Small group meetings with University attorneys, auditors, deans, the Faculty Senate Committee and Civil Liberties Board were held to better understand their perception of the issues that were critical in establishing the policy and to begin familiarizing them with the incidents that were occurring on campus.

3. *Defining the Role and Expectations of Policy and*

4. *Clarifying Community Standards and Values*

During these discussions, many values issues arose. They can best be described they represent different points on two key continua, the polarities of one are l and learning, the polarities of the other--individual responsibility and institutional responsibility.

The first key continua had polarities of law on end and learning on the other. One pole of this continuum of arguments represented the value of law and order. Some members of the community argued that no policy was needed in this area at all, that the laws were clear, that the institution had to be firm solid on these issues and do so without discussion, that these are matters of ri vs. wrong and that all violators should simply be referred to the law enforcement agencies and punished. Representatives of this pole argued against policy development in that they viewed policy as additional clutter to a system already in place. They argued that the institution is lax in enforcing existing laws, and that more needs to be done to prosecute offenders.

The other pole of this continuum represented the value of learning, experimentation and teaching. Representatives of this pole argued that policy was definitely needed, that although some laws exist to help define the boundaries of behaviors, the nature of information technology and its use causes people to not always see the value issues involved and sometimes causes them to do things before they know what they are doing. Most importantly for this group of people, policy was felt to be needed because Universities need to retail jurisprudence authority in order to provide the range of responses to a community of growing individuals, in order to teach individuals about the ramifications of their actions and not just punish them. Some argued that no matter what the violation or the seriousness of it, access to the information resource is the community's right and denying access would violate the bigger value by depriving the individual of the tools of their learning and so could never be justified.

The second key continuum had polarities of individual responsibility on one end and institutional responsibility on the other. Issues of Individual responsibility vs. institutional responsibility have and will be hotly debated a Michigan. The University of Michigan is a decentralized institution. Schools and Colleges are very autonomous in the handling of their affairs yet all are obligated to follow the institution's policies.

College students are a growing, developing group of individuals, may of whom are experiencing levels of responsibility for their own behavior for the first time. It is no surprise that a decentralized institution such as a University will struggle with and debate the parameters of a doctrine of individual responsibility. Naisbitt and Aberdeen write: "The first principle of the New Age movement is the doctrine of individual responsibility. It is a Westernized version of the ancient Eastern dogma of karma, that every action generates consequences the actor will eventually face. As the Bible puts it, "As you sow, so shall you reap." Individual responsibility, however, stresses the present; each individual is responsible for everything he or she does."....."This is not an "every man for himself" type of individualism, gratifying one's desires for their own sake and to hell with everyone else. It is an ethical philosophy that elevates the individual to the global level;...."4

When the institution puts in place a policy which includes words like "It is the policy of the University to maintain access to local, national and international sources of information.....It is the policy of the University that information resources will be used by members of its community with respect for the public trust.... "etc., the community asks who has the responsibility for providing this access, these resources-- the individuals, schools, colleges or the central administration? Do we imply with this policy that equal access to resources will be provided to all individuals of the community? Do we imply that all of the U- M regional campuses will also have equal access? Does the institution intend to take the responsibility for enforcement of the policy, or is this the responsibility of the individual? If an individual professor or student cannot afford or is no provided with the tools to do his/her work, and therefore is pressed toward unlawful copying of software, is this unethical behavior the institution's responsibility or the individual's responsibility? Those who argue for individual responsibility then are asked how far this responsibility goes. Does include "whistle blowing and reporting others who are violating the policy"?

While it seems clear that the institution has the responsibility for setting the standard, how much consensus of the community is required and how much involvement is needed? At Michigan we have taken the position that it is the institution's responsibility to involve the community in setting the standard an then take the lead in facilitating the discussion of ethics and values issues. Whether or not these facilitated discussions will be open, stimulating and productive in terms of movement towards Naisbitt and Aberdene's New Age individual responsibility, and towards a high level of social responsibility and moral behavior, remains to be seen. Struggling with these polarities and the polar values they represent: Law vs. Teaching and Individual Responsibility vs. Institutional Responsibility, is stimulating and very profitable work.

5. Gaining Administrative Endorsement and Promotion and

6. Expanding Scope to Include All Information Resources

In 1989 when the drafted policy was presented to the Executive Officers and Deans of the University, it was suggested that what had been drafted as a policy for information technology and network use, be expanded to include all information resources, technology and networks. This meant that members of our library community needed to consider the scope and effects of a policy of "proper use" in their area as well. After another six months of discussions and input from the library professionals, we defined information resources within this document to include "any information in electronic or audio-visual format or any hardware or software that make possible the storage and use of such information. Examples are electronic mail, local databases, externally accessed databases, CD-ROM, motion picture film, recorded magnetic media, photographs, and digitized information.

7. Gaining Executive Formal Approval

The policy, "Proper Use of Information Resources, Information Technology Resources and Networks at the University of Michigan", was approved in its expanded form in 1990, approximately two years after the University-wide Information Technology Policy Committee recognized the need, and approximately 3 years after the first major controversial incident. All Deans, Directors and Department Chairs in the University community of approximately 55,000 received a letter from the University Vice President encouraging them to study the policy and expressing the expectation that the policy would be implemented in each school, college and department at the University.

8. Information Dissemination and

9. "Thinking About It"-- The Policy Implementation Model

In 1991 a first stage implementation project was designed to stimulate information technology ethics and values discussions on the campus, to encourage people to become familiar with the policy and to encourage responsible citizenship in the use of the new technologies. This discussion process, is a very small beginning in what the author believes needs to be a much broader social awakening or massaging of values on this and other campuses.

The project was called the "Think About It" campaign. Volunteers were recruited as discussion facilitators from across the campus. Volunteers from a wide range of disciplines and responsibilities came forward. Interest in the to and

the issues was high. It was necessary to close the recruiting when the number of volunteers reached fifty.

Facilitators committed to participate in one, (four hour), training session and two, (one hour), facilitated discussions with a partner, wherever they were assigned on campus. They committed this time over the course of two semesters, Fall term 1990 and Winter term 1991.

Unlike most training programs, the volunteers were not provided with a set body of information which they in turn, would disseminate on the campus. They received background information regarding the "Proper Use" policy and information about significant existing state and national laws. However the primary thrust of the training and the project itself was to encourage people to think about the issues, to clarify their values, to learn to accept and evaluate other points-of-view, and to identify the ethics and values issues central to incidents and practices in the use of information technology.

The vignettes that were being collected from the start of the policy development work, were again critical to the process. Not only did they assist people who w developing the policy to become familiar with the range and scope of issues, but in the facilitator training, they served as case studies for developing and appreciating different points-of-view. Volunteers discussed a variety of differ incidents, evaluated the soundness of the policy against the incidents, determined whether or not the policy provided guidance, and finally, utilized the vignettes as discussion stimuli for the small group discussion they were invited to facilitate within the campus units.

The "Think About It" campaign was endorsed by the Executive Officers of the University. All Deans, Directors and Department Chairs were encouraged, by the Vice President, to invite facilitators for scheduled small group discussions wit faculty, staff and/or students within the schools, colleges and departments. Of the 50 trained facilitators, 33 were scheduled for small group sessions; some scheduled for one and others for as many as four sessions. A total of 31 differ groups were engaged in discussions. Group size ranged from 6 to 90. A total of approximately 620 people were involved in these discussions overall, during the period of October 1990 to March 1991.

The process promoted discussion and identification of many issues. The type of issues, as well as attitudes toward the issues, varied depending upon the area i which participants worked. For example, people most responsible for institutional data and the security of large systems, e.g. those in student record or financial systems such as payroll, had stronger opinions about the absoluteness of rules than did those more closely associated with the teaching and research duties of the institution. They also felt that exploration and experimentation with the technology was more unacceptable than did those in the academic areas, who often tended to value and encourage such behaviors. Defining what constituted "proper use", for particular areas and for particular kinds of transactions, was important work for the community.

According to facilitator ratings, approximately 80% of the participants commented favorably about the process, indicating that they enjoyed the opportunity to discuss and think about these issues, they gained clarification regarding issues and were able to get concerns into the open, they appreciated t opportunity to hear different points-of-view, and benefitted from the stimulation of this discussion at work. Approximately 90% of the facilitators rated this method of policy awareness and implementation positively. On a scale of 1 low to 4 high, 38% rated the method a (3), 14% a (3-4) and 38% a (4).

As one might expect, the biggest beneficiaries of this process were the facilitators themselves. 76% of the facilitators felt that the project had been very positive raising consciousness in the facilitator group, and 66% felt that the project h been very positive in raising issues for facilitators that will be discussed fur However, when the facilitators were asked how effective they felt the project ha been in raising consciousness for the campus community as a whole, they were conscious of the size of the community and the enormous task for which the project only provided a small effort of input. 60% of the facilitators felt that effectiveness of the project in raising campus consciousness regarding the issue was low to very low. 32% felt they could not access its effectiveness at all. felt that the project effectiveness in raising issues that will be discussed fur the campus as a whole was low to very low. 32% could not access its effectiveness in this area.

Therefore, while a very high percentage of the facilitators felt that this was a superior method for policy awareness and implementation, they also felt that it had a relatively small impact on the campus community as a whole. All of the facilitators felt that more needs to be done in this area and that this project should be viewed as only the first step in a long line of needed activities.

ISSUES FOR OTHERS

The following list provides a sample of the kinds of issues and questions that arise as information technology ethics policies are developed.

1. What constitutes wasteful practices as apposed to appropriate use of information technology resources?
2. Who takes responsibility for providing the resource, the hardware and software, if the institution defines in policy, and individual's responsibilities?-- the individual or the institution?
3. Who is responsible for reporting suspected violations of the resource if there is an ethics policy in place? Does this encourage vigilante behavior
4. Does the existence of a policy imply that the institution will take the responsibility to keep the staff current, up-to-date, and trained in the use of the resources?
5. What is the intended use of the resources? Is it only for work, learning, an research? Where is communication, experimentation, and personal enjoyment in this set of priorities?
6. What is authorized use and who authorizes it? What if the authorizer is unavailable and work must get done?
7. What resource access should be provided to fringe groups who want to participate, who are members of the community, who by their purpose are designed to harass or unsettle values and practices?
8. What other policies and regulations exist, either state or national, to which an institutional policy will have to be coordinated?
9. What effects will such a policy have on the practice of whistle blowing and what position will the university take when individuals confront supervisors for improper practices?
10. What is the balance between civil rights and civil liberties? What constitutes censored electronic freedom of speech and what, protection against harassment or intimidation?
11. What position does the institution want to take regarding open dialogue on university sponsored media? Is it different for private vs. public schools?
12. What is the line between policy and guidelines for a particular community?
13. Should different standards exist for faculty, staff, students?
14. Where should policy be placed within the institutions documents in order to effect the entire community? e.g. student disciplinary policies staff policy guides? etc.
15. What difference is there between using incidents as opportunities for teaching and learning, and being lax in enforcing ethical use of resources?
16. What constitutes malicious intent? Should that be spelled out in any policy and be the guiding principle for whether or not discipline occurs?
17. Are all processes within a policy domain protected with a right to grievance procedure and a protection of individual rights?
18. Who should have an investment, an ownership in the final policy? Who should want it implemented and see to it that it is?

SUMMARY

On University and college campuses, the wide variety of behavioral incidents arising in the use of information technology is creating a set of ethical and va questions that must be understood. They center around definition of responsible use of the technology.

When such incidents come to the attenti n of administrators within the institution, a poli_y is called for. For some, policy is meant as a control too unacceptable behavior. For others, a policy is a standard setting tool; it is t formal commitment of

the institution to a standard of behavior to which the community commits itself and to which it aspires.

Policy can go further than controlling and setting standards however. It can be tool through which other societal, developmental and institutional goals can be reached. It can be a vehicle for community involvement, one that stimulates discussions, that motivates the community to determine where it stands on particular ethical issues and how it wants to behave. It can be a dynamic document that not only provides the standards and the sanctions, but focuses on process, moving into arenas of learning and teaching and on to action and practice.

In 1987, stimulated by an incident of harassment on one of the student electronic conferences, the University of Michigan began to build a policy regarding ethical use of its information technology resources. Community involvement was basic to the process. In 1990, the policy called: "Proper Use of Information Resource Information Technology Resources and Networks at the University of Michigan" was approved for campus-wide implementation.

In 1991, a first stage implementation process, called the "Think About It" campaign, designed to stimulate ethics and values discussions campus-wide, was instituted using 50 faculty and staff volunteers as discussion facilitators. Approximately 31 different discussion sessions were held. Approximately 620 people directly participated in discussions of the ethics and values issues, debating proper use, and exchanged information about proper use practices.

No two communities will proceed through discussion of these ethical issues in the say way or at the same pace. It is clear that different issues will arise i different settings. It is clear as well that some of the issues that have arise the campuses that are most heavily invested and experienced in information technology use, are very likely to arise on other campuses as well, as use on the campuses increases. Thinking about these issues, understanding them for the basic values and ethics issues they contain, can prepare administrators for more open and knowledgeable responses. The promise of positive gains, of increasing community involvement, and of the massaging of social consciousness, is great.

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